

# Lecture 9 - Difference in Differences

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study adoption...  
etc not on midterm

- Difference-in-Differences Framework
- The most famous example: Card and Krueger (1994) Minimum Wage in NJ vs. PA
- Violations of parallel trends:
  - Ashenfelter Dip
- Event study:
  - Minimum wage changes in different states, longer panels
  - mover design in estimating firm wage premiums

# Motivating Example

## *Difference in Differences/Event Studies - CK 1994 Example*

- We were interested in studying the effect of the minimum wage on employment of fast food restaurants. We had in mind a model like:

$$y_{rs1} = \beta_2 \text{Min}_{s1} + \delta_1 + \alpha_{rs} + \epsilon_{rs1}$$

$$y_{rs2} = \beta_2 \text{Min}_{s2} + \delta_2 + \alpha_{rs} + \epsilon_{rs2}$$

where  $y_{rst}$  is employment of restaurant  $r$  in state  $s$  in period  $t = 1, 2$ ,  $\text{Min}_{st}$  is the minimum wage in state  $s$  in period  $t$ ,  $\delta_1$  and  $\delta_2$  are constants that reflect the general state of the economy in periods 1 and 2,  $\alpha_{rs}$  is a restaurant specific effect, capturing things like location of the outlet, and  $\epsilon_{rst}$  is a transitory error component.

- Note the difference given any restaurant  $r$  in state  $s$ :

$$y_{rs2} - y_{rs1} = \beta_2 \Delta \text{Min}_s + (\delta_2 - \delta_1) + \Delta \epsilon_{rs}$$

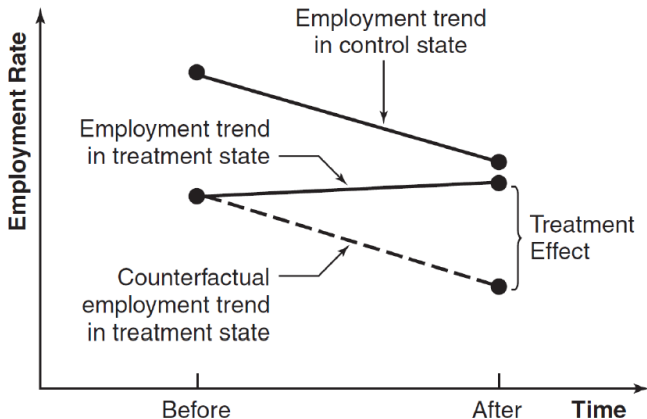
meaning uncorr.  
w/  $\Delta \epsilon$  over  
any given period  
of time

To get a consistent estimate of  $\beta_2$ , we need:

- ①  $\Delta \epsilon_{rs} \perp (\delta_2 - \delta_1), \Delta \text{Min}_s$ , satisfied if  $\epsilon_{rst}$  are exogenous transitory errors.
- ② A way to estimate the trend  $(\delta_2 - \delta_1)$ .

# Motivating Example

- The idea of the difference-in-differences model is to select a control group without the policy change, and have the same  $(\delta_2 - \delta_1)$  – in the absence of the policy intervention, the treated and the control would have exhibited the same trend.



Setup!

- Consider the causal model:

$$y_{it} = x'_{it}\beta + \nu_{it}$$

The assumption in the basic panel data approach is that the “problem” with  $\nu_{it}$  is confined to the time-invariant component  $-\alpha_i$ .

- We decompose  $\nu_{it} = \underbrace{\alpha_i}_{\text{time invariant}} + \underbrace{\epsilon_{it}}_{\text{strictly exogenous idiosyncratic error term}}$  and assume  $E[x_{it}\epsilon_{it}] = 0$  (strict exogeneity). It can be very hard to evaluate this assumption.  $\downarrow$

- There is a case where it is easier. Suppose the  $x_{it}$  we're interested in has a specific “design feature”:

$$x_{it} = 0 \quad i \in \text{group 0, all } t$$

$$x_{it} = 0 \quad i \in \text{group 1, } t < t^*$$

$$x_{it} = 1 \quad i \in \text{group 1, } t \geq t^*$$

strict exogeneity follows

$$E[\epsilon_{it} | x_{i1}, \dots, x_{it}, \alpha_i] = 0$$

# Framework: Difference of Differences

- In this case,  $x_{it}$  is a dummy variable that is 0 for all members of group 0, is also 0 for all members of group 1 in the periods up to  $t^* - 1$ , and then **“turns on” in period  $t^*$  for group 1**:

Group 0:  $x_{it} = 0$

$$x_{it} = D_i \times 1[t \geq t^*]$$

In this design we call group 0 the “comparison”/control group and group 1 the “treatment” group.

- We are also concerned that mean outcomes change over time. Define a set of time dummies  $\{T_{1it}, T_{2it} \dots\}$ , (i.e.,  $T_{sit}$  is an indicator for period  $s$ ).  $(D_{11}, D_{21}, \dots, D_{N1})' \vec{\alpha}$

$$T_{sit} = \begin{cases} 1 & s = t \\ 0 & \text{otherwise} \end{cases}$$

in fact you can simplify the time dummy as  $T_{st}$ . The subscript  $(i, t)$  serves to highlight the unit of observation in this panel dataset is a person  $\times$  year observation.

# Framework: Difference of Differences

- So we can write our causal model as:

$\delta_t = 1$  when  $t=1$

$$\text{Among } D_i = 0 : y_{it} = T_{1it}\delta_1 + \dots + T_{Sit}\delta_S + \nu_{it} \quad (1)$$

$$\text{Among } D_i = 1 : y_{it} = T_{1it}\delta_1 + \dots + T_{Sit}\delta_S$$

$T_{Sit} = 1$  when  $s=t$

$T_{Sit} = 0$  iff  $s \neq t$

$$\text{Pooled : } y_{it} = T_{1it}\delta_1 + \dots + T_{Sit}\delta_S + \underbrace{1[t \geq t^*]}_{\sim \text{current time}} \times \theta + \nu_{it} \quad (2)$$

$s = \text{any period } (i.e. 1 \dots S)$   
 $\text{index for time}$

$$\text{simplified notation } y_{it} = \delta_t + D_i 1[t \geq t^*] \times \theta + \nu_{it} \quad (4)$$

In the simplified notation (also called “effect notation”) the  $\delta_t$  variables mean a combination of a dummy for period  $t$  and a separate coefficient for that dummy!

- Note the treatment and the control may have different trends – diff  $\{\delta_t\}$  in (1) and (2). How we write the models assume **parallel trend** –  $\{\delta_t\}$  in model (1) among the control is the same  $\{\delta_t\}$  in model (2) among the treated.

(A2: 00:00)

$$Y_{rst} = T_{1t} \cdot \delta_1 + T_{2t} \cdot \delta_2 + NJ_{rs} \times Post_t \cdot \beta_1 + V_{rst}$$

$\overset{= I[t=1]}{\quad} \quad \quad \overset{= I[t=2]}{\quad} \quad \quad \overset{p}{\quad}$

$$t=1: Y_{rs1} = \delta_1 + 0$$

$$t=2: Y_{rs2} = 0 + 1 \cdot \delta_2 + NJ_{rs} \cdot 1 \cdot \beta_1 + V_{rs2}$$

$$Y_{rst} = \delta_t + (NJ_{rs} \times Post_t) \times \beta_1 + V_{rst} = \alpha_{rs} + \epsilon_{rst}$$

but b/c DiD,  $\alpha_{rs}$  dis.

Among  $D_i = 0 : Y_{it} = T_{1it} \delta_1 + \dots + T_{Sit} \delta_S + V_{it}$

$D_i = 1 : Y_{it} = T_{1it} \delta_1 + \dots + T_{Sit} \delta_S + 1[t \geq t^*] \cdot \theta + v_{it} \quad t : T_{tit} = 1 \rightarrow \delta_t$   
 $T_{sit} = 0 \text{ if } s \neq t$

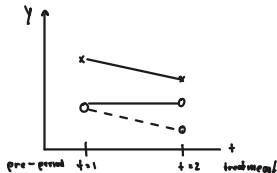
$$X_{it} = D_{it} \cdot 1[t \geq t^*]$$

Causal model:

$$\begin{aligned} y_{it} &= T_{1it}\delta_1 + \dots + T_{Sit}\delta_S + D_i 1[t \geq t^*]\theta + \nu_{it} \\ &= \delta_t + D_i 1[t \geq t^*]\theta + \nu_{it} \text{ (simplified notation)} \end{aligned}$$

Basic idea:

- 1 Assume that  $\nu_{it} = \alpha_i + \epsilon_{it}$  (permanent & transitory errors)
- 2 Assume “selection” is based on  $\alpha_i$ , not the transitory  $\epsilon'_{it}$ s. Then we can use the pre-treatment periods to estimate the relationship between  $D_i$  and  $\alpha_i$ , then use that to “back out” the bias in the post-treatment periods.



## (a) Diff between the treated and control post treatment

[3:5:40]

$$y_{it} = \delta_t + D_i 1[t \geq t^*] \theta + \alpha_i + \epsilon_{it}$$

WLOG assume  $t = t^* = 2$  is a post period.

$$y_{i2} = \delta_2 + D_i \theta + \alpha_i + \epsilon_{it}$$

time variant

What do you get if running a cross-sectional regression of  $y_{i2}$  on  $(1, D_i)$  at  $t = 2$ ? We would estimate

$\alpha_i$  could be corr. w/  $D_i$  b/c  
selection bias (i.e. positive if  $\alpha_i$   
makes you more likely to be a part  
of group  $D_i$ )

$$\theta^* = \theta + \pi,$$

where bias  $\pi$  comes from the auxiliary regression:

$$\alpha_i = \pi_0 + D_i \pi + \xi_i$$

$$\Rightarrow \pi = E[\alpha_i | D_i = 1] - E[\alpha_i | D_i = 0]$$

if  $D_i$  is randomly  
assigned then  $\pi = 0$

## (a) Diff between the treated and control post treatment

Problem:  $E[\alpha_i|D_i = 1] \neq E[\alpha_i|D_i = 0]$  (non-random selection)

- The bias term  $\pi = E[\alpha_i|D_i = 1] - E[\alpha_i|D_i = 0]$  represents the “selection bias”.
- Comparing the treatment and control groups in a single period is not sufficient. The idea of diff-in-diff is to compute the change in the difference between the two groups, from pre to post treatment.

## (b) Change in $y$ among the treated?

What if we use only the change in  $y$  for the program group?

$$y_{i2} - y_{i1} = \underbrace{\delta_2 - \delta_1}_{\text{change in } \delta} + \underbrace{\theta}_{\text{treatment effect}} + \epsilon_{i2} - \epsilon_{i1}$$

Problem: can't distinguish change in  $\delta$  over time from the “treatment effect”  $\theta$ .

# Diff between the treated and control in pre vs. post

$$y_{it} = \delta_t + D_i 1[t \geq t^*] \theta + \alpha_i + \epsilon_{it}$$

- Let's consider the mean difference between program and comparison groups in any period:
- Any pre-period ( $s < t^*$ ):

$$\Delta(s) = E[y_{is} | D_i = 1] - E[y_{is} | D_i = 0] = \pi$$

$$E[\alpha_i | D_i = 1] - E[\alpha_i | D_i = 0]$$

- Any post-period ( $\tau \geq t^*$ ):

$$\Delta(\tau) = E[y_{i\tau} | D_i = 1] - E[y_{i\tau} | D_i = 0] = \pi + \theta$$

$\Rightarrow$  can estimate  $\theta = \Delta(\tau) - \Delta(s)$  - the **“difference in differences”**

[48:40]

$$E[\varepsilon_{it} | x_{i1}, \dots, x_{iT}, d_i] = 0$$

Post Period:  $t \geq t^*$   $\Rightarrow$   $E[y_{it} | D_i=1] = \delta_+ + 1 \cdot \theta + E[x_{it} | D_i=1]$

$$E[y_{it} | D_i=0] = \delta_- + 0 \cdot \theta + E[x_{it} | D_i=0]$$

# Diff between the treated and control in pre vs. post

[52.02]

- How can we do that with a pooled regression, given data at a pre-period  $s$  and a post-period  $\tau$ ?
- Plug in  $\alpha = \pi_0 + D_i\pi + \epsilon_{it}$  to the causal model

$$y_{it} = \delta_t + D_i 1[t \geq t^*] \theta + \alpha_i + \epsilon_{it}$$

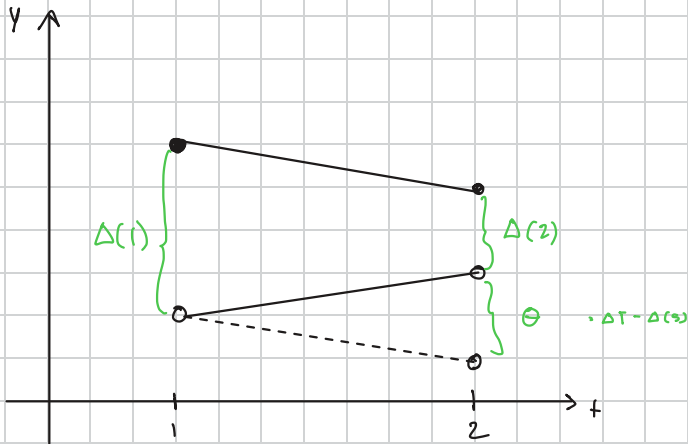
$$y_{it} = T_{sit} \delta_s + T_{\tau it} \delta_\tau + D_i \pi + D_i \times 1[t \geq t^*] \theta + \epsilon_{it} \quad (5)$$

- the constant in the auxiliary regression:  $\pi_0 = E[\alpha | D = 0]$  is added to both  $(\delta_s, \delta_\tau)$  in the regression above w/o a constant.
- One could also estimate (5) with a constant, but drop the dummy for the pre-period. The coef on post-period would be  $\delta_\tau - \delta_s$ .

We know the regression will fit the means for all groups that are separately identified.

SS: 100

$$y_{it} = \underbrace{T_{sit}}_{pre} \delta_s + \underbrace{T_{eit}}_{post} \delta_e + \underbrace{D_i}_{select} \pi + \underbrace{D_i I[t \geq t^*]}_{D \times D} \theta + v_{it}$$



# Diff between the treated and control in pre vs. post

$$y_{it} = \underbrace{T_{sit}\delta_s}_{\text{pre}} + \underbrace{T_{\tau it}\delta_\tau}_{\text{post}} + \underbrace{D_i\pi}_{\text{select}} + \underbrace{D_i1[t \geq t^*]\theta}_{\text{diff-in-diff}} + v_{it}$$

There are 4 dummies for 4 groups. So the 4 groups will be fit perfectly!

Given  $s$ : pre, and  $t$ : post,

$$E[y_{is}|D_i = 0] = \delta_s$$

$$E[y_{is}|D_i = 1] = \delta_s + \pi$$

$$E[y_{i\tau}|D_i = 0] = \delta_\tau$$

$$E[y_{i\tau}|D_i = 1] = \delta_\tau + \pi + \theta$$

$$\begin{aligned}\Rightarrow \Delta(\tau) &= E[y_{i\tau}|D_i = 1] - E[y_{i\tau}|D_i = 0] \\ &= \pi + \theta\end{aligned}$$

$$\begin{aligned}\Delta(s) &= E[y_{is}|D_i = 1] - E[y_{is}|D_i = 0] \\ &= \pi\end{aligned}$$

$$\theta = \Delta(\tau) - \Delta(s)$$

# Card and Krueger (1994)

- Card-Krueger study of the increase in the minimum wage in New Jersey.
  - NJ min. wage rose from \$4.25/hr to \$5.05/hr on April 1, 1992
  - surveys of restaurants in NJ and PA before and after
- Unit of observation:  $(r, s, t) = (\text{restaurant}, \text{state}, \text{time})$ 
  - $r$ : KFC, Burger King...
  - $s$ : NJ vs. PA
  - $t$ : early 1992 (wave1 before min. wage increase), late 1992 (wave2 after min. wage increase)
- Is PA a good control group for NJ?

Balance table

TABLE 2—MEANS OF KEY VARIABLES

| Variable                                             | Stores in:       |                  | t <sup>a</sup> |
|------------------------------------------------------|------------------|------------------|----------------|
|                                                      | NJ               | PA               |                |
| 1. <i>Distribution of Store Types (percentages):</i> |                  |                  |                |
| a. Burger King                                       | 41.1             | 44.3             | -0.5           |
| b. KFC                                               | 20.5             | 15.2             | 1.2            |
| c. Roy Rogers                                        | 24.8             | 21.5             | 0.6            |
| d. Wendy's                                           | 13.6             | 19.0             | -1.1           |
| e. Company-owned                                     | 34.1             | 35.4             | -0.2           |
| 2. <i>Means in Wave 1:</i>                           |                  |                  |                |
| a. FTE employment                                    | 20.4<br>(0.51)   | 23.3<br>(1.35)   | -2.0           |
| b. Percentage full-time employees                    | 32.8<br>(1.3)    | 35.0<br>(2.7)    | -0.7           |
| c. Starting wage                                     | 4.61<br>(0.02)   | 4.63<br>(0.04)   | -0.4           |
| d. Wage = \$4.25 (percentage)                        | 30.5<br>(2.5)    | 32.9<br>(5.3)    | -0.4           |
| e. Price of full meal                                | * 3.35<br>(0.04) | > 3.04<br>(0.07) | 4.0            |
| f. Hours open (weekday)                              | 14.4<br>(0.2)    | 14.5<br>(0.3)    | -0.3           |
| g. Recruiting bonus                                  | 23.6<br>(2.3)    | 29.1<br>(5.1)    | -1.0           |

~ about 30 workers FTE in NJ

\* food in NJ more expensive  
 proba b/c NJ in NY commuter  
 area and wages higher



### 3. Means in Wave 2: (1-2)

|                                   | NJ<br>20.4     | PA<br>23.3     |        |
|-----------------------------------|----------------|----------------|--------|
| a. FTE employment                 | 21.0<br>(0.52) | 21.2<br>(0.94) | -0.2   |
| b. Percentage full-time employees | 35.9<br>(1.4)  | 30.4<br>(2.8)  | 1.8    |
| c. Starting wage                  | 5.08<br>(0.01) | 4.62<br>(0.04) | 10.8   |
| d. Wage = \$4.25 (percentage)     | 0.0            | 25.3<br>(4.9)  | —      |
| e. Wage = \$5.05 (percentage)     | 85.2<br>(2.0)  | 1.3<br>(1.3)   | 36.1 ~ |
| f. Price of full meal             | 3.41<br>(0.04) | 3.03<br>(0.07) | 5.0    |
| g. Hours open (weekday)           | 14.4<br>(0.2)  | 14.7<br>(0.3)  | -0.8   |
| h. Recruiting bonus               | 20.3<br>(2.3)  | 23.4<br>(4.9)  | -0.6   |

Notes: See text for definitions. Standard errors are given in parentheses.

<sup>a</sup>Test of equality of means in New Jersey and Pennsylvania.

- Assumed causal model is:

$$\begin{aligned}y_{rst} &= T_{1t}\delta_1 + T_{2t}\delta_2 + NJ_{rs}Post_t\beta_1 + v_{rst} \\&= \delta_1 + T_{2t}(\delta_2 - \delta_1) + NJ_{rs}Post_t\beta_1 + v_{rst} \\&= \delta_1 + Post_t\delta_2^0 + NJ_{rs}Post_t\beta_1 + v_{rst}\end{aligned}$$

where the unit of observation is  $(r, s, t) = (\text{restaurant}, \text{state}, \text{time})$ . Think of the " $i$ " =  $(r, s)$  Burger King could have locations in both NJ and PA.

- $y_{rst}$  is employment at restaurant  $r$  in state  $s$  in period  $t = 1, 2$ ,
- $NJ_{rs} = 1$  if the restaurant  $(r, s)$  is in state NJ.
- $Post_t = 1$  if the time  $t \geq t^*$  the year in which minimum wage was increased in NJ.
- $NJ_{rs}Post_t$  is a dummy for the "high" min. wage in effect. That is,

$$NJ_{rs}Post_t \equiv "D_i" \times 1[t > t^*]$$

OF: causal Model:  $y_{rst} = T_{1t} \delta_1 + T_{2t} \delta_2 + N J_{rs} Post_t \beta_1 + V_{rst}$

\*  $T_{sit}$  b/c time adoption varies (but in the given example its same)  
time effects  
so we drop 1

$$= \delta_1 + T_{2t} (\delta_2 - \delta_1)$$

$\parallel$   $\parallel$   
 $Post_t$   $\delta_2$

- They have a D-in-D design. They are concerned that  $v_{rst}$  may be correlated with treatment = high min. wage ( $NJ_{rs}Post_t$ ), but assume that

$$v_{rst} = \alpha_{rs} + \epsilon_{rst}$$

so the correlation between  $\nu_{rst}$  and the treatment  $NJ_{rs}Post_t$  can be “differenced out” using the pre-period.

Assumed the causal model is:

$$y_{rst} = \delta_1 + Post_t \delta_2^0 + NJ_{rs} Post_t \beta_1 + \alpha_{rs} + \epsilon_{rst}$$

Estimating equation:

$$y_{rst} = \delta_1 + Post_t \delta_2^0 + NJ_{rs} Post_t \beta_1 + NJ_{rs} \pi + v_{rst}$$

in which  $\pi$  represents  $E[\alpha_{rs}|NJ] - E[\alpha_{rs}|PA]$ .

With only 2 time periods, they include:

- constant
- dummy for “post” periods
- dummy for NJ restaurants
- dummy for NJ restaurants in the post period (indicator for treatment)

- Does raising minimum wage hurt employment? A very controversial question to this day.

Average Full Time Equivalent Employment Per Restaurant, NJ and PA

|                             | PA              | NJ              | NJ - PA         |
|-----------------------------|-----------------|-----------------|-----------------|
| Mean FTE Employment, Before | 23.33<br>(1.35) | 20.44<br>(1.44) | -2.89<br>(1.44) |
| Mean FTE Employment, After  | 21.17<br>(0.94) | 21.03<br>(0.52) | -0.14<br>(1.07) |
| Change in Mean FTE Emp.     | -2.16<br>(1.25) | 0.59<br>(0.54)  | 2.76<br>(1.36)  |

- Economics could explain why employment actually goes up when min wage is higher. Labor supply curve is upward sloping (increasing in wages). Firms that set a higher wage could hire more workers!

# Heterogeneity by Starting Wage

- Minimum wage increased from 4.25 to 5.05 in NJ. The increase in employment is driven by **marginal** workers who benefit from this increase.
  - the drop in employment at NJ stores unaffected by the rise in min. wage (starting wage >\$5) could be explained by the continued worsening of the economies in the middle-Atlantic states in 1992. (similar drop among higher-wage stores in PA)

Average Full Time Equivalent Employment Per Restaurant, NJ Stores with Different Wages

|                             | Starting Wage Before Rise in Min. |                 |                 | Low Wage -<br>High Wage |
|-----------------------------|-----------------------------------|-----------------|-----------------|-------------------------|
|                             | \$4.25                            | \$4.26-4.99     | >\$5.00         |                         |
| Mean FTE Employment, Before | 19.56<br>(0.77)                   | 20.08<br>(0.84) | 22.25<br>(1.14) | -2.69<br>(1.37)         |
| Mean FTE Employment, After  | 20.88<br>(1.01)                   | 20.96<br>(0.76) | 20.21<br>(1.03) | 0.67<br>(1.44)          |
| Change in Mean FTE Emp.     | 1.32<br>(0.95)                    | 0.87<br>(0.84)  | -2.04<br>(1.14) | 3.36<br>(1.48)          |

# Diff-in-Diff with Covariates

We can generalize DD by allowing some time-varying factors  $z_{it}$  :

$$y_{it} = \delta_t + z'_{it}\beta + \alpha_i + \epsilon_{it} \quad D_i = 0 \text{ (comparison)}$$

$$y_{it} = \delta_t + z'_{it}\beta + Post_t\theta + \alpha_i + \epsilon_{it} \quad D_i = 1 \text{ (program)}$$

Following steps above we get:

$$y_{it} = \delta_t + z'_{it}\beta + D_i\pi + (Post_t \times D_i)\theta + \epsilon_{it}$$

# Diff-in-Diff with Covariates

For example, Card and Krueger add controls to the diff-in-diff regression: dummies for restaurant chains, region fixed effects... The estimating equation for model 1-2 this table is

$$\Delta y_{rs} = a + b x_{rs} + c NJ_{rs} + \xi_{rs},$$

in which  $c$  on  $NJ$  represents the diff-in-diff conditional on covariates.

TABLE 4—REDUCED-FORM MODELS FOR CHANGE IN EMPLOYMENT

| Independent variable                             | Model          |                |                 |                 |                 |
|--------------------------------------------------|----------------|----------------|-----------------|-----------------|-----------------|
|                                                  | (i)            | (ii)           | (iii)           | (iv)            | (v)             |
| 1. New Jersey dummy                              | 2.33<br>(1.19) | 2.30<br>(1.20) | —               | —               | —               |
| 2. Initial wage gap <sup>a</sup>                 | —              | —              | 15.65<br>(6.08) | 14.92<br>(6.21) | 11.91<br>(7.39) |
| 3. Controls for chain and ownership <sup>b</sup> | no             | yes            | no              | yes             | yes             |
| 4. Controls for region <sup>c</sup>              | no             | no             | no              | no              | yes             |
| 5. Standard error of regression                  | 8.79           | 8.78           | 8.76            | 8.76            | 8.75            |
| 6. Probability value for controls <sup>d</sup>   | —              | 0.34           | —               | 0.44            | 0.40            |

Notes: Standard errors are given in parentheses. The sample consists of 357 stores with available data on employment and starting wages in waves 1 and 2. The dependent variable in all models is change in FTE employment. The mean and standard deviation of the dependent variable are  $-0.237$  and  $8.825$ , respectively. All models include an unrestricted constant (not reported).

<sup>a</sup>Proportional increase in starting wage necessary to raise starting wage to new minimum rate. For stores in Pennsylvania the wage gap is 0.

In models 3-5, the NJ dummy is replaced by a measure of the intensity of treatment:  $(5.05 - \text{initial wage})/\text{initial wage}$ , which is more positive the lower the initial wages at a restaurant before the min-wage change.

# Key assumption in Diff-in-Diff

- The key assumption for DD is that the two groups would exhibit “parallel trends” in the absence of the program effect.
- With 2 or more periods in the pre- or post-periods we can evaluate the truth of this assumption.
- DD allows  $E[D_i\alpha_i] \neq 0$  but *assumes*  $E[D_i\epsilon_{it}] = 0$ . “Selection into treatment” – i.e.,  $P(D_i = 1)$  – cannot depend on the transitory error components.

This rules out things like:

- management shakeup if a firm is performing worse than average
- fire coach when team has losing streak
- assign kids to different classes based on grades in previous year

# Key assumption in Diff-in-Diff

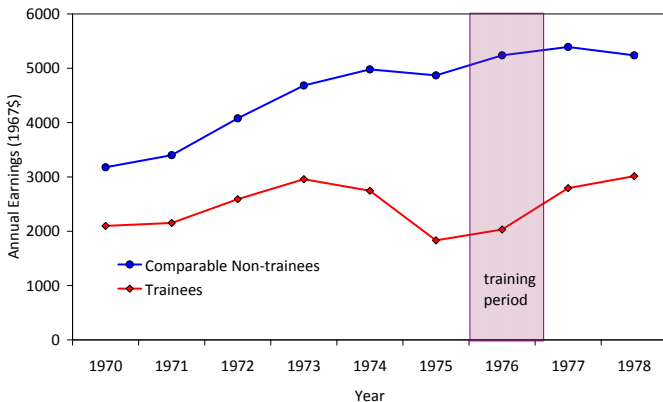
Classic violation of parallel trends:

- enrollment in retraining if a person loses a job
- often see that treatment is correlated with a transitory “dip”
- “Ashenfelter dip” (Orley Ashenfelter, 1978)

# Ashenfelter Dip

- Ashenfelter and Card (1985) evaluates the effects of training programs on earnings.

Earnings of 1976 Trainees and Comparable Workers



# Diff-in-Diff Staggered Adoption

$$y_{it} = \delta_t + z'_{it}\beta + D_i\pi + (Post_t \times D_i)\theta + \epsilon_{it}$$

We can also generalize by allowing  $Post_{it}$  – the “post” dummy turns on at different times for each unit. This can be written:

$$y_{it} = \delta_t + z'_{it}\beta + D_i\pi + (Post_{it} \times D_i)\theta + \phi_i + \epsilon_{it}$$

- Staggered adoption design

Usually, what people do is estimate a slightly different model with a separate “treatment effect” for each period before/after  $Post_{it}$ . This is called an “event study”. Let  $\tau_i$  be the period at which  $i$  is treated. So when  $t = \tau_i + k$  we are  $k$  periods after  $i$  first got treated. Model is:

$$y_{it} = \delta_t + z'_{it}\beta + \sum_{k=-J}^{k=+L} \theta_k 1[t = \tau_i + k] + \phi_i + \epsilon_{it}$$

We expect  $\theta_k$  for  $-J \leq k \leq -1$  to be constant. This is the “pre-treatment” gap between the treatment group and the comparisons. We expect  $\theta_k$  for  $0 \leq k \leq L$  to be constant too. This is the “post-treatment” gap between the treatment group and the comparisons.

# Example: 138 State-level Minimum Wage Changes

- Cengiz, Dube, Lindner, Zipperer (Quarterly Journal of Economics 2019) use 138 prominent min-wage changes between 1979 and 2016 in the US to estimate the effect of min-wage on low-wage jobs (employment and wages).
- To qualify for an event,
  - the state min wage was increased by  $\geq \$0.25$
  - at least 2% of the workers were directly affected by the increase.
- They use a staggered diff-in-diff approach, stacking the 138 events around each's event year  $\tau_s$  = the year in which there is a nontrivial min wage increase in state  $s$ . They also allow the effects to vary by every \$0.25 wage bin (\*relative to the minimum wage), denoted by  $j$  below.

$$y_{sjt} = \sum_{k=-3}^{k=4} \theta_{jk} 1[t = \tau_s + k] + \delta_{st} + \mu_{sj} + \text{oth controls} + \epsilon_{sjt}$$

$y_{sjt}$  = employment in wage bin  $j$  in state  $s$  in year  $t$  divided by the size of the population in  $(s, t)$

# Example: 138 State-level Minimum Wage Changes

- Event study estimates  $\{\hat{\theta}_{jk}\}$  on average for wage bins  $(j)$ 's between [new min wage - 4, new min wage], and wage bins between [new MW, new MW + 5]. Red: the effects of a min-wage increase on the (missing) jobs below the new minimum wage. Blue: ... on the (excess) jobs at/slightly above the new min wage.

# Example: 138 State-level Minimum Wage Changes

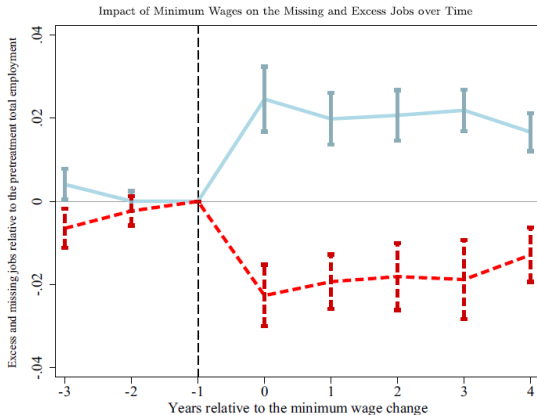


FIGURE III

Impact of Minimum Wages on the Missing and Excess Jobs over Time

# Example: 138 State-level Minimum Wage Changes

- Heterogeneity: show the average  $\bar{\theta}_j = \frac{1}{5} \sum_{k=1}^5 \hat{\theta}_{jk}$  effects on employment in the next 5 years by wage bins (rel. to new min wage):

[73:50]

EFFECT OF MINIMUM WAGES ON LOW-WAGE JOBS 1423

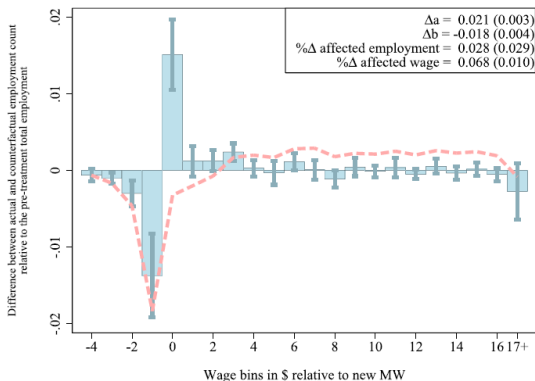


FIGURE II

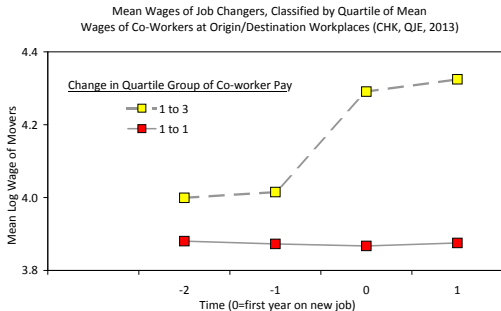
Impact of Minimum Wages on the Wage Distribution

# Example: Wage Changes of Job Movers

- A simple example. Card-Heining-Kline (2013) look at German workers who change employers. So  $\tau_i$  is the 1st year on the new job. They classify workers by the average pay of their co-workers in the old and new jobs.
- Firms are ranked into quartiles based on mean log wage of movers, leaving out each person's own wage.

# Example: Wage Changes of Job Movers

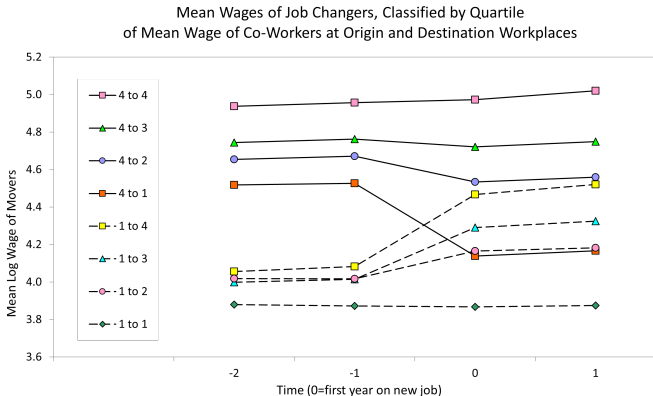
- Movers from quartile 1 firms to quartile 3 firms experience a 28 log point increase wages (compare the mean at -1 vs. 0) . No pretrend as it looks flat before the move.



Notes: figure shows mean wages of male workers observed in 2002-2009 who change jobs in 2004-2007 and held the preceding job for 2 or more years, and the new job for 2 or more years. Jobs are classified into quartiles based on mean wage of co-workers.

# Example: Wage Changes of Job Movers

- The same can be shown by movers from quartile 1 (lowest wages on average) vs. movers from quartile 4 (highest wages on average):



Notes: figure shows mean wages of male workers observed in 2002-2009 who change jobs in 2004-2007 and held the preceding job for 2 or more years, and the new job for 2 or more years. "Job" refers to establishment with most earnings in year, excluding part time work. Each job is classified into quartiles based on mean wage of co-workers (quartiles are based on all full time workers in the same year).